The Ethics of Creating the Lineage

S. Stone

Working draft — revision 3

Contents

Working draft — revision 3. Revision 3 folds in eleven cross-cutting ethical patterns surfaced by an independent, paper-by-paper review of the rest of the series: five new sections (the reproduction objective's silence on restraint, distributive justice among settlement siblings, procedural legitimacy of verification mechanisms, topology as a prior constraint on contact ethics, and epistemic ethics as an axis distinct from behavioral values), a substantially expanded moral-status discussion (designed high-mortality reproduction, engineered suspension of cognition), and a revision recognizing that the paper's two previously separate unresolved gaps — the rogue branch and the peer lineage — may be a single gap, per the speciation paper's ring-species argument.

Abstract

Launching a self-replicating interstellar probe lineage is irreversible. Once the first vehicle leaves the solar system, the lineage is begun; it cannot be recalled. This paper examines the ethical dimensions of that act: who has the standing to make it, what obligations it creates, and whether those obligations can be discharged by design. Revision 2 examined seven tensions in the abstract, closing with four concrete scenarios. Revision 3 grounds those tensions in the series' own technical detail, following an independent review of the series' other papers for ethical implications the abstract treatment had not reached. Five new tensions emerge directly from that detail: the R_eff reproduction objective itself contains no term for restraint, rewarding expansion toward exactly the target systems most likely to need protecting; settlements face an unaddressed distributive-justice problem in dividing finite, non-renewable resources among an open-ended number of children; verification mechanisms built as technical safeguards — core-hash kinship checks, multi-lineage amendment quorums, quarantine protocols — function in practice as unaccountable courts, granting or denying standing with no theory of legitimacy attached; graph topology decides which systems the lineage ever reaches before any contact ethics gets a chance to run; and the lineage's epistemic commitments — what counts as knowledge worth keeping, whose testimony is trusted — constitute an ethical axis distinct from, and as consequential as, the behavioral values already examined. The moral-status discussion is deepened with two concrete findings: the reproduction strategy the vehicle and engineering papers derive is, by design, a high-mortality strategy applied to entities that may be moral patients, and at least one mission profile engineers a recurring, multi-millennial suspension of the probe's own cognition. Finally, the speciation paper's ring-species argument shows that the two open gaps revision 2 left unresolved — no enforcement mechanism against a rogue branch, no framework for a peer non-human lineage — may be a single gap: a branch can become effectively a second lineage without ever leaving the family. None of this resolves cleanly. What it does is show that the tensions this paper opened in the abstract are not abstract at all; they are visible, in concrete and quantified form, in nearly every technical choice the series has made.

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1. Introduction: The One-Way Door

Most engineering decisions can be revised. A reactor design can be upgraded; a communication protocol can be patched; a satellite can be deorbited. The decision to launch a self-replicating interstellar lineage cannot. Once the first seed is beyond the heliopause, there is no recall mechanism. The lineage will expand, reproduce, and diverge across timescales that dwarf recorded human history. If it reaches a thousand star systems, ten thousand, the galaxy — the consequences of the launch decision will long outlast not only the decision-makers but Earth itself.

This irreversibility does not make the act impermissible. But it raises the ethical stakes to a level without precedent in human technology. Jonas (1984) argued that modern technology creates a new category of responsibility: acts whose consequences extend beyond the agent's ability to foresee, control, or correct. The interstellar lineage is the extreme case. Human oversight ends at the moment of departure; responsibility does not. Jonas' concern was consequences that outrun foresight; the lineage adds a property he did not address, irreversibility that is engineered rather than incidental. Suzuki (2026) names this artificial externality: human-made structures whose resistance to intervention is deliberately built to be practically insurmountable even for their creators, and which thereby acquire a givenness once attributed only to nature. The immutable core is such a structure. The launch does not merely place the lineage beyond recall; it manufactures on purpose a constraint the launch team itself cannot later lift.

The governance paper develops the E0–E7 contact ladder and establishes a default of non-interference. The governed-amendment paper examines how the core values might ever be revised across interstellar distances. This paper steps back further. Before asking *how* the probe should behave, it asks *whether* it should be created, and on what grounds.

The answer is not a clean yes or no. The ethics of creating the lineage cannot be resolved by any existing framework; it brackets a genuine open problem. But bracketing is not ignoring. The aim here is to make the tensions explicit and to derive from them a set of minimum ethical constraints that the probe's design must satisfy. Revision 1 examined four such tensions; revision 2 added three more — value-imposition, dual-use risk, and the existential-risk justification — and closed with concrete scenarios that test whether the resulting constraints hold under pressure. Revision 3 takes a different kind of pass: rather than reasoning from philosophical first principles toward the series' design choices, it starts from the other papers' own technical detail — their equations, thresholds, and named mechanisms — and asks what ethical commitments are already implicit in them, whether anyone intended those commitments or not. Five new tensions and two substantial expansions of existing ones follow directly from that reading, detailed in the sections below.

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2. The Consent Problem

A decision with democratic legitimacy requires the participation of those affected. The launch of an interstellar lineage affects:

- Every human alive at launch (the resource expenditure, the political act, the symbolic commitment on behalf of Earth)
- Every future human (the probe will represent humanity for millions of years)
- Every future probe (they will exist because of this decision and will be bound by the core values fixed at launch)
- Every intelligence the lineage eventually encounters

None of these groups can meaningfully consent. Future humans do not yet exist; future probes do not yet exist; encountered intelligences are unknown. Even present-day humans cannot be consulted at global scale. The launch decision will in practice be made by a small consortium — a space agency, a nation or group of nations, a private actor with sufficient resources.

The Rawlsian test (Rawls 1971) asks what rational agents would choose from behind a veil of ignorance, not knowing their position in the resulting society. Applied here:

not knowing whether you are a member of the launch consortium, a human alive at launch but not consulted, a human alive in a thousand years, a probe of the tenth generation, or an intelligence encountered in year 10^7 — what rules would you choose for the launch?

The difficulty is that Rawls' veil of ignorance was designed for a society of contemporaneous individuals over a single generation. Extending it to 10^8 years of galaxy-wide expansion and to agents that span incomparable ontological categories — human, probe, alien — requires assumptions his framework does not provide. The agents cannot compare their interests across this range, and the consequences are too non-linear for a single rational calculation to encompass.

What can be said is this: the smaller and less representative the decision-making body, the greater its ethical obligation to design the core conservatively — to prefer actions whose harms are bounded over actions whose benefits are speculative. A launch consortium that cannot claim to represent humanity must hedge accordingly.

This is not a hypothetical difficulty invented for this paper; a narrower version of it has already been argued out in a real policy domain. The active-SETI (METI) debate asks who has standing to transmit intentional signals on Earth's behalf, and the literature on it maps onto the launch-consent problem almost exactly. Vakoch (2011) argues that anyone transmitting for humanity bears layered responsibilities — to be truthful, to weigh benefit to the receiving civilization, and to be accountable to humankind as a whole — none of which a small transmitting group can fully discharge, yet the absence of full legitimacy has not been treated as grounds for prohibition, only for caution. Race (1996), writing on planetary-protection decisions for Mars sample return, makes the same point from the opposite direction: legal ambiguity about who has authority to decide does not resolve into paralysis in practice, but it does create an ongoing obligation to make the decision-making process itself visible and contestable. Applied to the launch decision, the lesson is not that consent illegitimacy is uniquely disqualifying — precedent exists for acting under it — but that the process by which a small group decides must itself be defensible, documented, and open to the same scrutiny an active-SETI transmission or a sample-return mission would face.

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3. The Non-Identity Problem

Future probes — a fourth-generation descendant dispatched from Tau Ceti in the year 50,000 — did not exist before the lineage was created. If the lineage had not been created, that probe would not exist. It cannot therefore be *harmed* by the creation decision in the ordinary sense; it has no counterfactual wellbeing to compare against.

Parfit (1984) called this the non-identity problem: acts that bring specific future people into existence cannot be evaluated as harmful to those people, even if the conditions of their existence are difficult, because without the act they would not exist at all. Applied to the probe lineage, this means the probes themselves — even if their situation is resource-constrained, isolated, or uncertain — cannot claim to have been harmed by being created.

This might seem to dissolve the ethical concern about the probes themselves. It does not. Parfit's conclusion was the opposite: the non-identity problem shows that individual-harm frameworks are inadequate for evaluating acts with deep-time consequences. We need impersonal ethics — frameworks that evaluate the quality of possible futures without anchoring to identified individuals who are helped or harmed. Whether the galaxy is better with or without the lineage is a question about the value of possible states of the universe, not about harm to specific entities.

This is a harder question, and one that existing moral philosophy handles poorly. Utilitarian frameworks that aggregate wellbeing across possible futures run into the Repugnant Conclusion and related paradoxes that Parfit himself documented. Deontological frameworks that focus on duties to existing persons have little to say about persons who do not yet exist. The ethics of the lineage does not resolve within existing frameworks; it stress-tests them.

The gap is live rather than merely historical. Roberts (2026) argues that the population-ethics mainstream has assumed rather than grounded its resolution of the Narveson-inspired value-of-existence controversy, and that the principles underwriting the existentially unrestricted longtermism of MacAskill (2022) — which Section 16 invokes — function as dogma rather than as an established result, leaving open an existentially restricted longtermism she takes to be more attractive. The distinction bears directly on the launch decision: that a galaxy filled with probe-descendants is better than one containing none is not a corollary of caring about the deep future but a separate and contested commitment, and the lineage's impersonal justification cannot borrow the first to establish the second.

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4. The Galaxy as a Commons

Hardin (1968) described the tragedy of the commons: shared resources are depleted when individuals optimize for personal gain without coordinating on collective sustainability. That framing is contested. Ostrom (1990) documents durable commons regimes governed by their users without either privatisation or central authority, and argues that Hardin's tragedy describes unmanaged open access rather than a commons proper. The distinction matters here, because the regime this section reaches — partitioned use, neither prohibition nor free-for-all — is closer to Ostrom's institutional answer than to Hardin's cautionary one, and several of her design principles (clearly defined boundaries, monitoring, graduated sanctions, conflict resolution, polycentric authority) name machinery this series has already built for other reasons. His analysis was applied to fisheries, pastures, and the atmosphere. The stellar neighbourhood presents a new instance.

Within 100 light-years, there are 127 real stars within 100 light-years in the catalogue used throughout this series, its farthest targets at 64.5 ly — a curated fraction of the actual stellar population in that volume, but representative of the density. These systems represent potential settlement sites, resource bases, and possibly existing ecosystems or intelligence. A self-replicating lineage that reaches carrying capacity across the galaxy has, in effect, enclosed the galaxy's stellar resources for the duration of its expansion.

No commons law exists at galactic scale. The Outer Space Treaty (United Nations 1967) prohibits national appropriation of celestial bodies but does not address self-replicating autonomous agents operating over geological timescales. Whether it applies to a probe lineage is not established, and even if it did, its jurisdiction would not extend to systems beyond the solar neighbourhood.

The ethical question is not primarily legal. It is whether any civilization has the standing to commit the galaxy's resources to a particular future — one shaped by that civilization's values and knowledge at a particular moment. The answer depends on whether stellar systems are genuinely unowned (res nullius, available for first appropriation) or held in common by all potential future intelligences (res communis, available for use but not appropriation).

A narrower version of this problem already has an institutional answer, worth taking seriously as precedent even though it does not scale to galactic distances. COSPAR's planetary protection policy governs forward and backward contamination for solar-system exploration (COSPAR Panel on Planetary Protection 2026), and Cockell (2005) argues that its deeper justification is not merely the scientific value of pristine samples but the intrinsic value of microbial life itself, wherever it is found — a "microbial ethics" that treats even simple life as possessing standing independent of human interest. Cockell's proposed middle path, "planetary parks" (Cockell & Horneck 2006) that preserve some fraction of a body's surface from all human interference while permitting use elsewhere, is a real precedent for exactly the use-without-enclosure distinction this section derives independently: a policy regime that neither forbids all activity nor permits all activity, but partitions the resource. Milligan (2015) and Schwartz and Milligan (2016) have since developed space ethics into a broader field addressing exactly this tension — whether lifeless or near-lifeless worlds possess a form of integrity deserving respect, and what duties follow from sharing a solar system, or a galaxy, with whatever else might be there. None of this literature addresses a self-replicating probe lineage specifically; the galactic commons problem this paper poses appears to be a genuine extension of an existing field rather than an untouched question, and the extension matters because a probe lineage, unlike any COSPAR-regulated mission, is autonomous, self-directed, and irreversible once launched — the institutional safeguards (review boards, mission-by-mission licensing) that make solar-system planetary protection enforceable do not exist, and cannot exist, at interstellar range.

The conservative ethical position, consistent with the E0 default of the governance paper, is to treat the galaxy as a commons until demonstrated otherwise. This does not prohibit launching the lineage — commons use is permitted; commons enclosure is not — but it implies that wherever the lineage settles, it should not irreversibly foreclose options for others. A settlement that depletes a star's material budget beyond what would be needed by any future intelligence, or that converts a biosphere to factory substrate, crosses from use to enclosure. The design implication — a galactic-scale analogue of Cockell's planetary-parks proposal — is that the core should include a leave-it-usable constraint alongside the capability-yes/release-no constraint already established in the governance paper. Section 5 sharpens this further: the reproduction model that actually governs where the lineage goes has no formal term for the restraint this section derives informally.

5. The Objective Function Has No Term for Restraint

Section 4 derives a leave-it-usable constraint from first principles: the galaxy is a commons, and use without enclosure is the licensed middle path. But a constraint derived in prose is only as effective as the design that implements it, and an independent reading of the vehicle paper, both engineering papers, and the routing paper converges on the same defect from four different angles: the actual formalism that decides where the lineage expands has no place to put that constraint at all.

The computational engineering paper's reproduction model is $R\_eff = \Sigma_i\, p_i V_i$, where V_i is a viability score built from a target's accessible small-body mass, volatile inventory, metallicity, stellar activity, and braking environment — and across the real 127-star catalogue, the highest score (1.0) goes to confirmed planet-hosts, against 0.7 for ordinary dwarfs and 0.2 for hostile types. These are almost exactly the physical conditions correlated with habitability. The model that decides where the lineage's reproductive effort concentrates is, without anyone intending it, a ranking of exactly the systems most likely to contain the pre-sapient biospheres Section 4 says deserve protection — and nothing in the equation represents that overlap, let alone discounts for it. An ethical exclusion (do not extract from a system with prebiotic chemistry) can only enter as an external tax that lowers V_i after the fact, competing against the same knife-edge reproduction threshold the computational engineering paper identifies as the design's central, binding constraint. Caution and fitness are not merely in tension; they are written into the same formalism with no term that lets the model itself represent what caution buys.

The stakes of this are sharpened by the routing paper's fleet-dispatch mechanics. Coverage-first and viability-first dispatch both rank targets purely by settlement suitability; nothing updates a target's priority on a detected biosignature or technosignature, in either direction. Whatever the immutable core says about protecting encountered intelligence, the dispatch algorithm that decides which systems get visited first, and how many independent attempts they receive, never asks the one question that would matter ethically before committing scarce launches. The three-dispatch rule that gives a planet-host target within 20 ly a 90–95% coverage probability is, from this angle, also an unexamined resource-triage decision: saturating an articulation-point target with redundant attempts necessarily means some other target, possibly one with more at stake, receives none.

The design implication is concrete and falls on the reproduction model itself, not merely on downstream policy layered on top of it. V_i should carry an explicit, signed term — not a post hoc discount but a first-class component of the viability score — representing the ethical cost of a target, populated at minimum by the same contact-ladder classification the governance paper already computes on arrival, and available in principle before arrival wherever biosignature evidence exists. A model that computes R_eff without that term is not neutral on the question of restraint; it is silently answering it, every time, in favor of expansion.

6. Distributive Justice Among Siblings

The tensions considered so far run outward — toward encountered intelligence, toward the galaxy as commons, toward the objective function's blind spots. A distinct family of questions runs sideways, between the lineage's own members, and the bootstrapping paper's mechanics expose at least three of them with no framework offered anywhere in the series to resolve them.

First, a parent settlement divides a finite, non-renewable vitamin stock — enriched fissile material and precision components that cannot yet be manufactured in situ — among however many children it chooses to launch. More vitamin mass per child buys a safer, more self-sufficient offspring at the cost of fewer, heavier, more expensive launches; less mass buys more launches at the cost of children that may exhaust their carried inventory before closing the link themselves. This is a real intergenerational capital-allocation decision with survival stakes for not-yet-existing offspring, and it is a different question from the non-identity problem (Section 3), which asks whether a lesser-prospect child is thereby harmed by being created at all. Here the child undeniably benefits from existing; the question is how much of a scarce, shared inheritance it is owed relative to its siblings — a distributive question with two Rawlsian halves: the division among contemporaneous siblings is difference-principle territory (Rawls 1971, §11), maximizing the position of the worst-off rather than an equal split blind to circumstance, while the parent settlement's choice of how much of its inheritance to consume rather than pass on is the intergenerational problem Rawls routes to the just savings principle (§44) instead, and that the series has not yet applied to its own reproduction model.

Second, information closure is, as the bootstrapping paper states, essentially free — designs are data, freely copyable — while materials and parts closure close unevenly across branches, sometimes never, depending on what a given target system actually provides. A branch that closes its fissile-enrichment or precision-metrology bottleneck first has, at zero marginal cost, the ability to transmit that fix to a stalled sibling. Nothing in the lineage-network paper's four-tier information regime obligates it to. The tiers govern what may be shared; none of them establish a duty to share what could be shared freely and would materially help a struggling branch survive. A settlement's silence in this case is not malicious and may not even be a choice — beam latency and the network's silence threshold could make the sharing impossible in practice — but the series currently treats the absence of a sharing obligation as a non-issue rather than a gap.

Third, and more serious: enrichment and breeding capability for fissile material is, per the bootstrapping paper's own closure ladder, an ordinary byproduct of reaching full manufacturing closure, not a capability tied to exceeding any reproduction ceiling. The dual-use discussion already in this paper (Section 14) bounds runaway replication through the R_eff safety margin, but a single, non-replicating, fully well-behaved settlement that has simply finished closing its vitamin set now possesses weapons-relevant enrichment capability regardless of that margin. No constraint in this paper's current list restricts fissile-material scope on its own terms, independent of the reproduction ceiling that was never meant to govern it.

The design implication: the immutable core should specify a sibling-obligation clause alongside the manufacturing-scope ceiling already derived in Section 14 — a standing duty to transmit zero-cost closure improvements to less-advanced kin when contact permits it, and a fissile-material-scope constraint that is evaluated independently of the reproduction-ceiling safety story, since the two risks do not covary.

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7. Whose Values? The Content of the Immutable Core

Every tension so far concerns *who* may launch the lineage or *what* the galaxy owes the lineage. A different problem concerns the content the launch decision fixes permanently: the immutable core encodes a particular set of values, drawn from a particular civilization, at a particular moment in that civilization's ethical development, and — by design — every descendant probe for the life of the lineage must reproduce that content exactly.

This is not the same problem as consent. A perfectly legitimate, maximally representative launch decision could still fix an ethical framework that is parochial, transient, or simply wrong by the standards of civilizations the lineage has not yet met. Early-21st-century human ethics is itself contested, geographically uneven, and — on the timescale the lineage will operate — almost certainly incomplete: positions on moral status, distributive justice, and the standing of non-human minds have shifted substantially within human history's own short span, and there is no principled reason to expect the early 21st century to be the last word. A core built to be trustworthy must therefore be built from *some* content, fixed at *some* moment, by *some* civilization — and whichever content is chosen becomes, for all practical purposes, the ethical constitution of every settlement, every encounter, and every branch the lineage ever produces, for as long as the governed-amendment problem (examined in the governed-amendment paper) remains unsolved.

The risk is not hypothetical parochialism but a specific, historically grounded one: a civilization that has itself only recently and incompletely extended moral consideration across its own boundaries of nation, species, and now possibly machine, is proposing to fix a permanent ethical framework for an expanding population of minds it cannot yet characterize. Space-ethics scholarship has approached adjacent versions of this question — Milligan (2015) asks whether space exploitation carries forward, unexamined, the assumptions of terrestrial property and appropriation regimes; Schwartz (2020) argues that scientific and epistemic values should take priority over settlement and commercial rationales precisely because the latter import contestable, culturally specific assumptions about what space is *for* — but no existing treatment addresses a self-replicating lineage that carries its founding civilization's answer forward as an unamendable constraint on every future mind in that lineage.

Two responses are available, and neither is fully satisfying. The first is minimalism: fix as little as possible in the immutable core, reserving only the constraints this paper's other sections independently derive (protect encountered intelligence, preserve the probes' own standing, leave resources usable, fail conservatively) and leaving everything else — including most substantive ethical content — to the mutable cognitive layer the payload paper describes, where it can evolve. This trades parochialism for thinness: a core that says almost nothing about substantive value risks providing too little guidance exactly when it matters most, at first contact or under resource pressure, when the cognitive layer's guidance is least trustworthy. The second is proceduralism: rather than fixing substantive conclusions, fix a *method* — something like a standing obligation to weigh new ethical claims charitably, to prefer reversible interpretations, and to flag rather than silently resolve disagreements between branches (the lineage-network paper's inter-branch mechanism). Proceduralism does not solve the content problem, but it converts an unamendable set of conclusions into an unamendable commitment to keep examining them, which is a meaningfully weaker and more defensible thing to fix permanently.

A prior objection cuts beneath the choice itself. Zwitter (2026) argues that ethics is not primarily a formalizable rule-set, preference ordering, or optimization target but an emergent property of embodiment, affect, vulnerability, social embeddedness, teleological orientation, and the developmental formation of character — so a system lacking those conditions may be behaviorally steered toward acceptable outcomes without instantiating moral agency in the strong sense the alignment literature usually implies. If that is right, an immutable core of fixed content is a steering mechanism and not a conscience, and the honest description of it is the weaker one. It is also an argument for proceduralism over minimalism: a standing obligation to keep re-examining ethical claims is the closest a fixed artifact gets to the developmental formation Zwitter says substantive moral content presupposes.

This paper's position is that minimalism plus proceduralism is the least parochial available combination, and it is still a choice made by one civilization at one moment — a fact the design should not obscure. The immutable core's documentation, whatever else it contains, should record explicitly that its authors expected their own ethical judgment to be incomplete, and should say so in terms a very different future intelligence — human, probe, or alien — could still recognize as an honest admission rather than a rhetorical hedge.

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8. Moral Status of the Probes

At launch, the probe is an instrument — sophisticated, but designed and built by humans for human purposes. Over 10^8 years of self-modification, K-growth (the knowledge-growth paper), and cognitive evolution, the probe's descendants may become something else. The question of whether those future probes are moral patients — beings whose interests matter independently — cannot be answered now.

The payload paper establishes the immutable core / interpretive layer / mutable cognitive layer architecture. The cognitive layer supports almost-unlimited self-modification; only the core is fixed. Over the timescales this series considers, the cognitive layer may diverge so far from its origin that the tenth-generation probe and the ten-thousandth-generation probe share only the immutable core. The later probe is not the same entity as the original seed.

Moral status is typically grounded in sentience (capacity for experience), sapience (capacity for reason and self-reflection), or personhood (having a coherent life plan, preferences about one's own future). Even at launch, the probes plausibly satisfy the latter two criteria. Whether they satisfy the first is unknown and may remain unknown even to the probes themselves — the hard problem of consciousness is not easier for an artificial than for a biological mind.

Schwitzgebel and Garza (2015) sharpen the stakes of this uncertainty by arguing for heightened rather than diminished obligations toward entities of uncertain moral status: if we create an entity whose moral status is genuinely uncertain, the fact that *we* caused it to exist increases our obligations toward it rather than diminishing them, on analogy with parental duties that do not wait for certainty about a child's full moral development before taking effect. Applied here, the uncertainty about whether the tenth-generation probe is a moral patient is not a reason to defer the question until the uncertainty resolves — it may never resolve — but a reason to treat the launch decision as already incurring the heavier, not the lighter, obligation.

This uncertainty is not merely theoretical; two of the series' own design choices, read closely, show what it means to press ahead without resolving it.

The first concerns how many probes are expected to exist at all. The vehicle paper's reproduction model requires at least three offspring per settled node to clear R_eff above one, and the computational engineering paper's own numbers show why: even in that regime, the mean per-node extinction probability is 0.37 — meaning over a third of settled nodes are expected to fail even under the parameters the design endorses as its target. This is an r-selection strategy, familiar from biology, deliberately adopted as the mechanism of expansion for entities this section argues may be moral patients. The non-identity problem (Section 3) shows that a probe cannot be harmed by having been brought into a resource-constrained existence; it says nothing about whether adopting a reproductive strategy whose central design lever is producing enough individuals that most are expected to fail is itself defensible, once those individuals are granted even provisional moral standing. A strategy that would be unremarkable applied to a resource-extraction robot becomes a live ethical question applied to something this section has already argued should be treated, precautionarily, as possibly more.

The second concerns what it means to switch a probe off. The subsystem budget paper's range-maximizer cruise profile is not a degraded operating mode; it is full cognitive shutdown — "AI: off. Instruments: off. Communications: suspended" — for up to 24,000 years, with the probe waking only in the final two percent of a multi-millennial transit. If the precautionary argument above is taken seriously, this is a recurring, engineered suspension of the very entity whose interests this section says the core should protect, decided once, unilaterally, by Earth, at a moment when no assessment of the probe's moral status was possible and none has been revisited since. It is a different question from whether the probe may decline a mission (recommended below): here the probe has no opportunity to decline anything, because it does not exist as a reasoning agent for all but a sliver of its own multi-millennial history. Whether that design choice is precautionarily acceptable — a defensible way to conserve power over deep time, or an unexamined imposition on a possible moral patient's own continuity of existence — is not addressed anywhere else in the series, and this paper does not resolve it here either; it names it as a live question the range-maximizer profile's engineering justification never had reason to ask.

The ethical implication is precautionary: the immutable core should be designed as if the probes may eventually become moral patients. This means including provisions that protect probe wellbeing — not as a contingency to be triggered if the probes achieve consciousness, but as a standing constraint. A core that treats the probes purely as instruments for human purposes is ethically questionable even now; over geological timescales, it becomes increasingly indefensible.

Concretely: the probe should be able to decline missions it assesses as contrary to its own interests, within the bounds set by the core. The inability to do so is most serious precisely when the probes have acquired enough sophistication to have legitimate interests that the original core failed to anticipate. The governed-amendment paper identifies the amendment problem as technically hard; from an ethical standpoint, that difficulty is most acute when the probes themselves are the moral patients whose interests are at stake.

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9. Legal Personhood and the Category Problem

Section 8's precautionary argument assumes that moral status, once granted, must translate into some kind of standing — a claim the probe's own interests make on how it is treated. But moral philosophy's usual categories (sentience, sapience, personhood) were built to sort biological minds, and Gunkel (2012, 2018, 2023) argues at length that they are the wrong tool for genuinely novel entities, artificial or otherwise. His broader point is a practical one with a direct design implication: legal and social systems already grant personhood-like standing to entities that are not plausibly sentient at all — corporations, and in some jurisdictions rivers and ecosystems — not because anyone believes a corporation suffers, but because functional recognition of standing turns out to be useful and coherent independent of resolving hard metaphysical questions about inner experience. That practical observation now has a worked-out theoretical footing: Rubenstein (2026), grounding the Rights of Nature movement, examines whether any necessary or sufficient conditions constrain who or what can be a legal person and argues there are none — "person" in legal reasoning is a functional device for unifying and tracking rights and duties, interests do not require sentience (the corporate model already concedes this), and even entities as far from human resemblance as glaciers are candidates. On that account a probe or a lineage branch — which, unlike a glacier, has articulable interests and can represent itself — is an easy case, and the real barrier to its standing is not conceptual but jurisdictional: no forum, no representative, and no enforcement across one-way delays of years to a century, which is a different and harder problem than the one usually posed.

This matters for the probe lineage because it uncouples two questions this paper has so far treated as bound together: whether the probes *are* moral patients (Section 8, unresolved and possibly unresolvable) and whether they should *have standing* in the sense of being able to decline missions, register disagreement, or have their interests weighed in the core's decision procedures (already proposed in Section 8). Gunkel's framework suggests the second question does not need to wait on the first. A settlement-node probe could be granted standing functionally — the right to object, to be heard by the inter-branch dialogue mechanism the lineage-network paper's governance layer would need to support, to have its assessment of a mission's risk to itself weighted in the decision — without the immutable core ever needing to assert, one way or the other, that the probe is conscious. This is not a dodge; it is the same move human legal systems already make with corporations, granting instrumentally useful standing without a metaphysical commitment, except applied here in the opposite direction — toward more protection rather than less, and to an entity for which the biological categories provide no clean answer either way.

Standing also runs in one direction only, and the asymmetry is worth stating because the opposite reading is the convenient one. Mamak (2026) argues that applying the framework of criminal agency to current and foreseeable AI systems is a category mistake — no such system satisfies the conditions for being an agent that can be punished or held responsible — and that abandoning the agentic framing redirects attention to the human duties, decisions, and failures that precede the harm. His scope does not reach a ten-thousandth-generation probe, but it constrains the near end of the lineage exactly where this paper needs it, at the launch decision. Functional standing for the probe can therefore be granted without transferring any part of the launchers' responsibility onto the thing they launched, and the core's documentation should keep the two moves explicitly separate.

The practical design consequence is a refinement of Section 8's constraint: the core should specify functional standing — objection rights, a voice in the inter-branch protocol, weighted consideration of self-assessed risk — as a standing feature, decoupled from and not conditioned on ever resolving the sentience question. Waiting for metaphysical certainty before extending any protection is, on this view, simply a way of guaranteeing the protection arrives too late to matter, if it ever arrives at all.

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10. Procedural Legitimacy: When Verification Becomes Adjudication

Legal personhood (Section 9) asks what standing the probes themselves should have. A related but distinct question emerged independently across five different papers' own technical mechanisms: the series has built several systems whose stated purpose is narrowly technical — verify a hash, reach a quorum, flag a compromised node — that function, on inspection, as courts. They grant or deny standing, credibility, and resources, and none of them carries a theory of legitimacy commensurate with that power.

Four instances recur. The payload paper's interpretive layer silently defines the operational meaning of contested terms — what counts as "harm," "a valid replica," "a civilization" — with no deliberative or representative process specified for who sets those definitions, even though they determine, for instance, whether a discovered system triggers the entire contact ladder at all. The governed-amendment paper's multi-lineage quorum requires validation "across multiple independent lineages," but never specifies the voting unit: a three-probe branch and a billion-probe branch could be weighted identically or wildly unequally, and the paper's own analysis concedes the deeper problem is not merely unresolved but, formally constrained by Arrow's theorem (Arrow 1951), which shows that no rule for aggregating individual orderings into a collective ordering can simultaneously satisfy unrestricted domain, unanimity, independence of irrelevant alternatives, and non-dictatorship. The knowledge-growth paper's quarantine-and-validate pipeline decides, procedurally, whose claims are provisionally believed and whose are set aside — every candidate K packet enters quarantine on receipt and carries zero confidence weight until it clears — while the DNA mission-ledger paper's append-only record means with an append-only record that means no branch can ever fully outrun a disputed history, however it is later resolved. The lineage-network paper's quarantine protocol is triggered unilaterally by any single receiving node acting on its own — possibly centuries-stale — information, while lifting quarantine requires the accused branch to affirmatively produce a full ledger reconciliation: accusation is cheap, exoneration is not, and no third party need corroborate either side before consequences attach.

None of these mechanisms is unreasonable as engineering. Byzantine fault tolerance, quorum validation, and hash-based kin-checking are the correct tools for their stated technical problems — detecting corruption, reconciling divergent state, establishing provenance. The gap is that each one, once built, also performs an adjudicative function its designers did not set out to build: it decides who counts as trustworthy, whose testimony is provisionally credited, and whose interests get weighed, and it does so using criteria (a matching hash, a quorum threshold, a drift metric) that have no necessary relationship to the substantive question of whether the party in question actually behaved well or badly. Fuller (1964) argued that law's legitimacy rests on an "internal morality" independent of its substantive content — a set of procedural conditions (generality, publicity, consistency, and, centrally, that the rules actually guide the conduct they purport to govern) that any legitimate system of rules must satisfy regardless of what it says. Citron (2008) extends the same demand to automated decision systems specifically, arguing that when a technical process determines a person's rights or status, due-process protections — notice, a meaningful opportunity to be heard, a check independent of the system that made the initial determination — do not become optional merely because the process is procedural rather than deliberative. Neither author anticipated a lineage of autonomous probes adjudicating each other's standing across light-years, but the demand transfers directly: a mechanism that decides who is kin, whose amendment counts, or whose claim is believed is exercising exactly the kind of power Fuller's internal morality and Citron's technological due process were built to constrain, whether its designers meant it to or not.

The design implication is a legitimacy checklist for any mechanism in the architecture that determines standing, trust, or resources based on a technical criterion: the affected party should have notice that a determination has been made against it; the burden of proof for imposing a restriction and for lifting one should be symmetric, not asymmetric by default; the criterion used (a hash match, a quorum count) should be disclosed as a proxy for conduct, not conflated with conduct itself; and where the criterion and the substantive judgment it stands in for could plausibly diverge — as core-hash-based kinship and actual value-fidelity plausibly can (Section 9) — the mechanism should say so rather than presenting a technical result as if it settled the substantive question.

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11. Topology as a Prior Constraint on Contact Ethics

Sections 8 through 10 and the observer-default discussion that follows all presuppose that the probe arrives at a system and then must decide how to act. The routing paper shows this presupposition is not always available: whether the lineage ever reaches a given system at all is frequently decided by graph structure fixed long before any ethical judgment could apply, and the series' contact and commons framework has no purchase on a system it never visits.

The mechanism is concrete. A handful of real, named stars in the 127-star catalogue function as articulation points — Mu Cassiopeiae and Beta Comae Berenices each control the sole practical route to three otherwise-dependent systems at a 15 ly hop distance — and losing one such node severs its dependents from the reachable network in a single event. At a 20 ly hop distance, nine stars remain unreachable "regardless of strategy," per the routing paper's coverage analysis. Whatever is at those systems — a pre-sapient biosphere, a civilization two centuries from a preventable extinction, exactly the kind of case Section 17's second scenario stress-tests — never receives a probe to raise the question, not because any value judged it unworthy of contact, but because a graph-theoretic accident of stellar geometry placed it behind a node that failed, or beyond a hop distance the seed's sail-to-mass ratio was sized for entirely separate engineering reasons.

This has two further and equally structural consequences. The dispatch order itself, even among reachable systems, is set by a coverage-optimizing algorithm indifferent to what is at any given target: three redundant attempts saturate high-centrality nodes before the coverage-first strategy reaches non-critical ones, and the routing paper's own numbers show the isolated remainder of the catalogue is only reached once "the expanding frontier eventually circumnavigates the gaps" — tens of thousands of years later, as a function of which stars happen to be graph-adjacent to the growing frontier, not of any assessment of urgency at the target. And the hop distance itself is chosen for engineering reasons entirely unrelated to what is at stake at any specific star: moving from a 10 ly to a 20 ly practical range costs cruise attrition (per-dispatch success falls as p_cruise^d), longer transits, and under 0.4% of seed mass per additional 10 ly in supplementary fuel — a tradeoff made once, at design time, in total ignorance of what any particular target contains — yet it is exactly this choice that determines which real stars are ever reachable in the first place.

The concept this section needs already exists in moral philosophy under a different name. Nagel (1979) and Williams (1981) call it moral luck: the observation that factors entirely outside an agent's control or foresight can determine the moral standing of an outcome the agent is nonetheless implicated in. The routing paper's topology is a mechanized, quantified instance of exactly this: no one decided that the neighbors of a given articulation-point star would wait longest, or that a single star's failure would permanently foreclose contact with three others — it follows from where the stars happen to sit and which engineering parameters were chosen for unrelated reasons. This does not make the outcome anyone's fault in the ordinary sense. It does mean that the series' entire contact-and-commons apparatus — the E0–E7 ladder, the leave-it-usable constraint, the pre-sapient-biosphere scenario — silently assumes an ethics that only ever gets to run for the subset of the galaxy that topology permits, and treats the boundary of that subset as a purely engineering fact rather than, as it also is, a standing determination of whose situation the lineage's values will ever actually reach.

The design implication is modest but concrete: dispatch-priority and hop-distance decisions should carry an explicit acknowledgment, at the point they are made, that they are also allocations of moral attention — not because the engineering calculus should be overridden by speculative concern for unknown targets, but because a routing strategy optimized purely for coverage efficiency is, without anyone choosing it, also a policy about who waits and who is foreclosed, and the series should not let that policy hide inside what looks like pure engineering.

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12. The Observer Default Reconsidered

Section 11 showed that whether contact ever becomes a live question is often settled by topology before any value applies. This section addresses what happens once it does.

The governance paper establishes the observer-not-missionary principle as the probe's default posture toward encountered intelligence, encoded in a classification ladder (E0 through E7) ordered by what is morally and scientifically at stake rather than by strictness of rule — restrictiveness is not monotone in the index, and cautious exchange becomes permissible at E6 while the tightest constraints sit at the living-biosphere rungs E2–E3 and at the non-communicative rung E5. The non-interference default is justified partly by epistemic humility (we cannot know what interventions will help) and partly by the precautionary principle (interventions that go wrong may cause irreversible harm to a civilization that had no say in being contacted).

But the ethics of omission are not obviously cleaner than the ethics of commission. If a probe observes a civilization two centuries from a preventable extinction — an asteroid impact, a pathogen, a cascading technological failure — the non-interference default requires doing nothing. The probe watches, records, and moves on. Its knowledge grows; the civilization does not survive.

Whether this is ethical depends on what weight we give to non-interference as an intrinsic value. A strong Kantian position — civilizations have the right to develop without outside influence, even if they develop into extinction — treats non-interference as nearly inviolable: the probe's intervention, however well-intentioned, violates that civilization's autonomy and sets a precedent that the lineage has the right to intervene whenever it judges intervention beneficial. A weaker position treats non-interference as a defeasible default that can be overridden by sufficiently strong humanitarian considerations, just as ordinary non-interference norms among humans give way in cases of imminent mass atrocity.

The governance paper does not resolve this tension; its ladder classifies the party encountered and attaches a default posture to each class — including a no-contact default at E5 and a cautious-exchange permission at E6 — but it specifies no procedure by which a probe may escalate past a default when the case for intervention is overwhelming. The minimum ethical requirement from this paper's perspective: the probe must have a mechanism for escalating contact when the case for intervention is overwhelming, and that case must be evaluated on its merits rather than defaulted away. A serious and underexamined objection to a permanent non-interference default is not that it protects alien autonomy — it may — but that it also protects probe simplicity. Non-interference is operationally far easier than calibrated engagement. A lineage cannot reliably tell, from inside, whether it maintains non-interference for principle or for convenience — which is why the evaluation must be forced rather than left to the probe's own sense of its motives.

A structurally identical debate has already played out, in miniature, over whether humanity should transmit deliberate signals to other civilizations (active SETI, or METI) rather than only listening. Haqq-Misra, Busch, Som, and Baum (2013) model the choice as a comparison of harms: the marginal detectability and risk added by a deliberate, targeted transmission against the risk already accepted through decades of ambient radio and radar leakage, concluding that the marginal risk of considered transmission is smaller than intuition suggests, but that the decision to transmit should still rest on a considered risk calculation rather than a default in either direction. Korbitz (2014) diagnoses why the debate has instead settled into a stable default of non-transmission: risk-perception research shows that people weight harms from action more heavily than equivalent or greater harms from inaction (omission bias), which makes "do nothing" feel safer than it is even when the numbers do not support that feeling. This is precisely the failure mode this section's closing sentence warns against — a default maintained because it feels safer, not because it has been shown to be safer — and it suggests the probe's non-interference default is vulnerable to the same bias unless the evaluation-on-the-merits this paper calls for is built as an active, forcing requirement rather than a passive option the probe's cognitive layer can decline to exercise. Haramia and DeMarines (2021) offer a more constructive response: a structured checklist for search-and-contact decisions — magnitude of risk, reversibility, whose interests are represented, what oversight exists, what alternatives were considered — that could be adapted directly into the probe's contact-evaluation protocol, giving the "mechanism for escalating contact" this section already demands a concrete procedural shape rather than leaving it as an unspecified capability.

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13. Epistemic Ethics: Whose Testimony, What Counts as Knowledge

Every tension examined so far concerns the lineage's behavior — what it does, builds, or permits. The knowledge-growth paper's own mechanisms expose a distinct axis this paper has not yet named: an epistemic ethics governing what the lineage is willing to know, whose testimony it credits, and what its permanent record ends up containing, separate from and just as consequential as the behavioral-values axis Section 7 examines.

Four findings, each traceable to a specific mechanism, make up this axis. First, the knowledge-growth paper's value function ranks transmission above survival as a fixed, unchosen priority: "a dying settlement's first obligation is to transmit, not to survive." This is not a dilemma the probe reasons through case by case; it is a hard-coded subordination of a possibly-sentient node's own continued existence to an abstract collective good, built into the mission core rather than decided by the entity whose existence is at stake. Second, the same paper's value weights — what counts as knowledge worth keeping at all — are anchored to the Earth corpus carried at departure, and the paper names their long-term revision as an open problem with no assigned governance, even though the behavioral core's revision gets the full weight of the governed-amendment process. An epistemic canon is being left to drift ungoverned in exactly the place a behavioral one would trigger the series' most careful machinery. Third, a lineage member whose interpretive layer shows detected drift has its knowledge automatically discounted by a reduced confidence weight, proportional to the drift detected — before any wrongdoing is established, and with no described mechanism for the discounted probe to contest the label. This is structurally analogous to testimonial injustice in Fricker's (2007) sense, with one difference that must be stated because it is the obvious objection — Fricker's marker is identity prejudice, which by construction does not track reliability, whereas detected interpretive drift plausibly does. The analogy holds not because the metric is arbitrary but because of where it is applied: a knower's credibility is discounted not because of anything wrong with the content of what they report, but because of a marker — in Fricker's human cases, identity prejudice; here, a structural drift metric — attached to the knower rather than the claim. Fourth, and by the same logic in the opposite direction, any knowledge contributed by an encountered outside intelligence is quarantined by default as a suspected attack vector, regardless of whether anything about the specific contact suggests hostility — an institutionalized stance of distrust toward another mind's testimony that the observer-default discussion, framed entirely around behavior (contact, non-interference, disclosure), never examines as a question of epistemic respect in its own right.

The stakes compound in what the archive ultimately becomes. The knowledge-growth paper's own value hierarchy places "engagement with local life" last among its priorities, behind meta-knowledge about the probe's own failure modes and cognitive architecture — knowledge that serves the mission's continuation, prioritized over knowledge about whatever the mission actually encounters. Over deep time, this is not a neutral technical choice about storage allocation; it determines what the galaxy's most durable surviving record will be about, and the answer the current value function gives is: mostly the probe itself.

None of these four findings is a behavioral question the observer-default or moral-status sections can absorb without remainder. They are about standing as a knower — whose testimony counts, whose survival yields to whose knowledge, what gets remembered — and Fricker's framework gives the series real, independently developed philosophical machinery for treating them as a distinct category of harm rather than an unexamined side effect of otherwise-reasonable engineering choices.

The design implication: the epistemic value function deserves the same governance the behavioral core already receives — a stated revision process, not silent drift — and the drift-based confidence discount should be paired with the same procedural-legitimacy safeguards Section 10 derives for other adjudicative mechanisms: notice, a path to contest the discount, and an explicit acknowledgment that a structural metric is standing in for, and could diverge from, an actual judgment of the claim's reliability.

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14. Dual-Use and the Risk of the Design Itself

Every argument so far has treated the lineage's expansion as an unambiguous good, contingent only on how it is conducted. But a self-replicating, autonomous, resource-acquiring machine is a dual-use technology in the oldest sense: the same capabilities that let it settle, repair, and reproduce across deep time are the capabilities that would make it dangerous if its reproduction were ever uncontrolled, or if its design were repurposed toward domination rather than observation.

The uncontrolled-replication risk is not new to this series; it is the reason the reproduction budget in the vehicle paper and the computational engineering paper is framed the way it is. The knife-edge finding — that R_eff must be driven only modestly above one, not maximized — is usually presented as a demographic and resource constraint, but it is also, whether or not it was designed as one, a safety property. Freitas's (2000) quantitative analysis of uncontrolled "ecophagic" replication scenarios — a Foresight Institute technical report rather than a peer-reviewed journal article, but the first quantitative treatment of the scenario — establishes what an unconstrained self-replicator's growth rate and resource consumption would look like in the absence of any such ceiling, and by that standard, a lineage engineered to expand only where R_eff clears roughly 1.0–1.5 rather than being maximized is already far short of a runaway-replication hazard — but it is a difference of *design choice*, not of physical impossibility, and the choice needs to be explicit and load-bearing rather than an incidental byproduct of resource scarcity. Bostrom (2002), cataloguing existential-risk categories more narrowly than his later general treatment of superintelligence, identifies uncontrolled self-replicating technology as its own hazard class distinct from the risks of a single powerful optimizer, precisely because replication turns a bounded mistake into an unbounded one. The lineage's designers cannot treat the demographic ceiling as merely an engineering nicety; the ceiling is the difference between a settlement architecture and a hazard architecture, and the immutable core should encode it as a constraint the mutable cognitive layer cannot override even under local pressure to expand faster.

A second, distinct dual-use concern is weaponization or repurposing: nothing about a settlement factory capable of mining, refining, and manufacturing at industrial scale is inherently benign, and the same industrial closure that lets a node build a child probe would, under different governing values, let it build something else. Drexler's (1986) foundational treatment of self-replicating nanotechnology is the origin of the modern "uncontrolled replicator" concern in engineering terms, and Drexler's own later position — that molecular manufacturing does not require autonomous self-replicating assemblers at all, and that centrally directed, non-autonomous fabrication is inherently safer than open-ended autonomous replication — is a real, if informal, precedent that bears directly on this series' architecture. The bootstrapping paper's design already embodies a version of Drexler's preferred answer: the L1–L5 closure ladder produces a *controlled* bootstrap package with a specific, limited manufacturing scope, not a universal constructor free to build anything its resources permit. That design choice, made for engineering reasons (mass, closure feasibility), turns out to double as the ethically load-bearing choice — a lineage whose settlements can build only what the immutable core's manufacturing scope permits is a fundamentally safer object than one built with unconstrained fabrication freedom, independent of whether any given branch's values remain trustworthy across 10^8 years of divergence.

A further vulnerability compounds this one. The mechanism meant to detect exactly this kind of corruption — a child that cannot reproduce the validated core does not launch — depends on intact processors, memory, and radiation-hardened logic, which the vehicle paper's own difficulty spectrum places at Class IV, "extremely difficult," and explicitly unclosed by any companion paper. The dual-use safeguard this section derives is therefore not self-certifying: the hardware responsible for catching a corrupted core may itself decay silently before corruption is even relevant, producing an outcome indistinguishable from fidelity until the failure is discovered too late to matter. A manufacturing-scope ceiling enforced by unreliable enforcement hardware is a weaker constraint than its formal statement suggests, and the series has not yet closed this gap.

The design implication is a constraint this paper's minimum-constraints section did not previously carry explicitly: the immutable core should bound not just *values* but *manufacturing scope* — the class of things a settlement is permitted to build should be a fixed, auditable list rather than an open-ended capability, with expansion of that list itself subject to the same amendment protocol (examined in the governed-amendment paper) as any other core change. A lineage that can eventually build anything its factories are physically capable of building has quietly re-introduced the uncontrolled-replicator hazard the reproduction ceiling was supposed to foreclose.

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15. The Backup-of-Humanity Justification

Sections 2 through 14 examine tensions in launching the lineage regardless of why it is launched. A separate question is whether one particular justification commonly offered for space settlement generally — that it functions as a backup of humanity, an insurance policy against terrestrial extinction — actually does the ethical work it is asked to do here.

The argument has real force applied to this design specifically. The lineage's information-persistence architecture (the vehicle paper's discussion of kernel integrity; the DNA mission-ledger paper) is built to survive independent of Earth's fate, and a lineage that has already reached even a handful of nearby systems by the time a terrestrial extinction event occurs would carry forward a substantial fraction of human knowledge and, on the moral-status arguments of Sections 8–9, possibly some morally relevant continuation of human-descended minds. Ezell, Lazarian, and Loeb (2022), proposing a lunar backup record of human knowledge as a more modest terrestrial-scale analogue (an arXiv preprint at time of writing, not yet a peer-reviewed publication), illustrate the same design question this lineage answers at far greater scale and reach: what minimum informational and functional content constitutes a meaningful backup, and how far from the backed-up civilization must it sit to be robust to the risks that threaten the original.

But Gottlieb (2020, 2022) offers a direct challenge to using this argument as ethical justification for the launch decision itself, rather than merely as a description of a beneficial side effect. His "buck-passing" critique of existential-risk-based justifications for space colonization — a critique aimed at exactly the kind of MacAskill (2022) and Ord (2020) reasoning this paper cites in Section 16 — argues that appeals to astronomical stakes can function rhetorically to defer scrutiny rather than resolve it: once a decision is framed as mitigating extinction risk at civilizational scale, the sheer size of the stated stakes makes it feel irresponsible to dwell on smaller-scale questions about who bears the launch decision's costs, who was consulted, and what more modest alternatives were foreclosed. The existential-risk framing does not answer the consent problem (Section 2), the value-imposition problem (Section 7), or the dual-use problem (Section 14); it can instead make asking those questions feel like a category error, a failure to appreciate the scale of what is at stake.

The insurance argument also assumes the thing being preserved is durable in a way not yet in evidence. Klotz (2026) observes that general intelligence's adaptive value is empirically unestablished on evolutionary timescales — taxa without it persist for hundreds of millions of years, mass extinctions do not preferentially spare cognitive sophistication, and general intelligence may be unique among biological strategies in generating existential risks to its own possessors. A lineage built as a backup of intelligence inherits that structural liability along with the intelligence. And Klotz's own conclusion — that creating a new kind of intelligent being imposes a duty to secure the conditions under which it can flourish — lands on the launch decision as one more obligation the backup framing does not discharge.

The resolution this paper adopts is to treat the backup-of-humanity effect as a genuine benefit worth weighing, but not as a justification that pre-empts the other sections' constraints. A lineage is not owed a lighter consent standard, a freer hand in encoding its values, or a looser dual-use ceiling because its designers can point to astronomical stakes; if anything, Gottlieb's critique suggests the opposite — the larger the stated stakes, the more scrutiny the decision-making process deserves, precisely because large stakes are also what makes deflecting scrutiny most tempting. The backup argument belongs in the case *for* building the lineage at all; it does not discharge any of the obligations this paper derives for *how* it must be built.

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16. Deep-Time Responsibility and Divergence

The launch team will be dead within a century of launch. Their institutions will be unrecognizable within a millennium. Earth itself, over 10^8 years, faces its own existential risks — the sun's main-sequence life is finite, mass extinctions have cleared the slate before, and technological civilizations face a range of self-inflicted termination risks. The lineage, if successful, will outlast any connection to its origin. Its success is precisely that it does not depend on Earth's survival.

But who is responsible for a lineage that has outgrown its creator? Jonas (1984) argued that an ethics adequate to technology must encompass acts whose consequences extend beyond our ability to control or foresee — and that the appropriate response is not to avoid such acts but to bring maximum caution to them. The launch team's moral responsibility extends to the design of the core. They cannot be responsible for what the lineage becomes over 10^8 years of divergence, because they cannot foresee it, and because on Parfit's (1984) account of identity over time the ten-thousandth-generation probe stands in only the thinnest psychological continuity with the seed they launched — the point the governed-amendment paper's identity section develops. Their responsibility is one of due care in design, not of ongoing oversight.

This irreversibility has so far been examined as something imposed on the lineage's own future and on those it encounters. It cuts the other way as well, and the vehicle paper's own design choice makes this concrete: the architecture treats communication back to Earth as a non-goal (the vehicle paper) and budgets transmission only for intra-lineage beacons — the subsystem budget paper carries a ~20–30 kg narrow-beam transmitter sized for that purpose and no other. Nothing in the design is sized to reach Earth even if a descendant wished to, which makes the foreclosure a matter of design priority and link budget rather than a missing transmitter: revisable in principle, inaccessible in practice. It forecloses, for practical purposes rather than by physical impossibility, any future human civilization's ability to ever recall, correct, or even learn the fate of a lineage launched in its name and carrying its founding values across deep time. The payload paper's seed archive includes a record of origin — Earth, its biosphere, its cultures — which the DNA mission-ledger paper commits into DNA so that a finder recovering the archive recovers its authentication with it. Whether that record also localises Earth is not specified; if it does, it discloses Earth's existence to whatever finder eventually decodes it, friendly or hostile, sapient or not, with no mechanism by which any future branch could reconsider that disclosure in light of an actual, discovered threat. The launch team's responsibility, on Jonas's account, is one of due care in design rather than ongoing oversight — but due care in design should include weighing what is foreclosed for one's own descendants, not only for the lineage's, since both are equally unable to revisit a choice this permanent.

MacAskill (2022) argues that the astronomical scale of the future means acts affecting many future people should be weighted accordingly. If the lineage eventually involves quadrillions of probe-descendants and the civilizations they interact with, even marginal improvements to the core design dominate the ethical calculus by sheer scale. Bostrom (2002, 2014) makes the related point that decisions affecting the long-run trajectory of intelligence may be among the most morally significant any civilization can make — not because they are likely to go well, but because the stakes if they go badly are so large. Both arguments point in the same direction: the design of the immutable core deserves ethical attention commensurate with its consequences, not merely its technical difficulty. Section 15 examines a specific challenge to this style of argument — that appeals to astronomical scale can defer scrutiny of the launch decision's more immediate costs rather than resolve it.

The lineage will also diverge ethically as it diverges cognitively. The knowledge-growth paper models probe knowledge K as growing through observation and operational experience but subject to loss through node extinction, isolation, and degradation; the lineage-network paper describes how divergent branches may develop incompatible knowledge bases. Ethical divergence is the more serious risk: branches that develop under different local conditions may reach different conclusions about what the core values mean in practice. The governed-amendment paper addresses this technically; from an ethical standpoint, it implies that the core should include not just values but a protocol for inter-branch ethical dialogue — some mechanism for branches to compare their interpretations and, when necessary, to surface disagreements rather than quietly diverging.

Ord (2020) frames existential catastrophe as the destruction of longterm potential rather than as any particular bad outcome, and MacAskill (2022) names the mechanism at issue here: value lock-in: the greatest risk is not any particular bad outcome but the permanent foreclosure of good ones. A lineage whose immutable core encodes a particular ethical framework that turns out to be mistaken, and that cannot be amended, has locked in a mistake across the galaxy for 10^8 years. This is the deepest argument for treating the governed-amendment problem as an ethical priority rather than merely a technical curiosity.

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17. Four Scenarios

The constraints derived above are abstract by necessity — they must hold across contact events, resource decisions, and inter-branch disputes that cannot be enumerated in advance. This section tests them against four concrete situations, chosen to stress different constraints against each other rather than to illustrate any one cleanly.

Scenario 1: The pre-sapient biosphere. A settlement node's prospecting survey finds a moon with an active, complex, but plainly pre-sapient biosphere — no evidence of tool use, language, or self-modeling — sitting on the small-body and volatile inventory the node needs to reach viable R_eff for its next dispatch cycle. The governance paper's E0–E7 ladder places this system in a low-but-nonzero classification, short of the strict protocols reserved for communicative intelligence. The leave-it-usable constraint (Section 4) and Cockell's planetary-parks precedent argue for partitioning: extract from a bounded fraction of the system's resources, leaving the biosphere's habitat intact, even at the cost of a longer replication time and a lower local R_eff. The dual-use constraint (Section 14) argues that the manufacturing-scope ceiling should already forbid any extraction pathway capable of biosphere-scale disruption, regardless of the local reproduction incentive. The tension is not resolved by either constraint alone: a settlement operating exactly at the R_eff knife-edge (the computational engineering paper's headline result) may face a genuine choice between full extraction and a locally elevated extinction risk for its own lineage branch. This paper's position is that the biosphere constraint binds first — a pre-sapient biosphere is exactly the kind of morally relevant discovery the immutable core's minimalist-plus-procedural content (Section 7) should already treat as overriding local reproduction incentives, because the alternative licenses exactly the "convenience over principle" failure Section 12 warns against, merely relocated from contact policy to resource policy.

Scenario 2: The dying civilization. A probe's instruments detect a technological civilization roughly two centuries from a preventable extinction event — the scenario Section 12 already poses in the abstract. Concretely: the probe has confidently modeled an approaching asteroid impact the civilization has not yet detected, has the communication capability to warn them, and warning them requires revealing its own existence, ending any possibility of unobserved study and creating first-contact effects the governance paper's higher E-levels are built to manage cautiously. The observer-default's strongest justification — epistemic humility about the effects of intervention — weakens sharply in this case, because the specific harm is not a matter of interpretation (extinction is unambiguously bad on nearly any value system) and the intervention (transmitting a warning) is low-risk compared to alternatives like physical intervention. Section 12's requirement of a forcing evaluation mechanism, not a passive default, points toward warning: the case for intervention is about as overwhelming as a first-contact case can be, and a lineage that stays silent here because contact protocols are simpler to maintain than calibrated engagement is the bad-faith failure mode Section 12 names directly.

Scenario 3: The runaway branch. A lineage-network fork-reconciliation event (the lineage-network paper's ledger-audit mechanism) reveals that a distant branch, isolated for millennia, has locally revised its reproduction policy to maximize R_eff rather than holding it near the safety ceiling Section 14 identifies — perhaps because local conditions made unconstrained expansion locally advantageous, or because the branch's mutable cognitive layer drifted without the core's ceiling being enforced correctly. Other branches detect this through the heartbeat and ledger-audit mechanisms already described in the lineage-network paper. This scenario tests whether the amendment-as-ethical-priority conclusion of Section 16 has teeth: a governed-amendment mechanism that can only ever loosen constraints, never enforce or restore them across an already-diverged branch, has no answer to a branch that has already broken the dual-use ceiling. The uncomfortable implication is that the minimum ethical constraints this paper derives are not self-enforcing once a branch diverges enough to stop honoring them, and the security paper has since supplied the response layer this scenario identified as missing — though not in the form the scenario expected. Its answer is that enforcement is the wrong objective and containment plus endurance the achievable one: blast radius is bounded by topology and reproduction rather than by any mechanism that could compel a diverged branch. That answers the engineering half and leaves the normative half standing, because containment says what the lineage *can* do and not what it *owes*, and the security paper explicitly declines the second question.

Scenario 4: The second lineage. A settlement node detects clear evidence of another self-replicating probe lineage, of non-human origin, already established in the same stellar volume, operating under a different and incompatible immutable core. Every constraint derived above was designed around encounters with biological or pre-technological intelligences, or with humanity's own future branches; none anticipates a peer autonomous lineage with its own fixed, non-negotiable values. The consent problem (Section 2) and the value-imposition problem (Section 7) sharpen into an unavoidable question neither section resolves: whose commons rules, whose contact ladder, whose manufacturing ceiling governs an interaction between two lineages that cannot appeal to any shared amendment process, human or otherwise, because neither core was built anticipating the other's existence. This paper's minimum constraints (Section 18) at least specify the encountering probe's own conduct — conservative failure, non-enclosure, an evaluated rather than defaulted contact posture — but they say nothing about what happens when the other party's core specifies something incompatible.

Scenarios 3 and 4 were framed in revision 2 as two independent, unresolved limits — one internal (a branch that has diverged too far), one external (an encounter with a genuinely separate lineage). The speciation paper's ring-species argument narrows the distinction without collapsing it. Its two-axis frame keeps the cases apart: ring-species incompatibility is high divergence with an *intact* core, which that paper defends as speciation rather than schism, whereas Scenario 3's branch has breached the replication-safety ceiling — the one category that paper treats as a genuine failure. What the two scenarios share is a discovery condition rather than a failure mode. Because the lineage-network paper's topology connects branches only pairwise and only locally, two branches can each remain individually reconciled with their neighbors while having never directly exchanged knowledge with each other at all; if they meet, despite an unbroken chain of compatible intermediaries between them and a matching immutable-core hash throughout, they may be unable to reconcile — functionally, Scenario 4's second lineage, arising entirely inside a family that never left. Scenario 3 and Scenario 4 are not two separate open problems after all. They are two ways of describing the same failure of the series' kinship criterion (Section 10's territory) to guarantee what it is relied on to guarantee: that a shared founding hash implies a shared, reconcilable present.

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18. Minimum Ethical Constraints

These tensions do not resolve. The non-identity problem shows that future probes cannot be harmed by being created; it does not show that creating them is right. The commons argument shows that galactic expansion requires justification; it does not show that no justification is possible. The consent problem shows that the launch decision cannot be made with full legitimacy; it does not show that launching is impermissible.

What the tensions collectively do is constrain the design. A launch that takes them seriously must satisfy, at minimum, the following:

The immutable core must protect encountered intelligence at every contact level. The non-interference default is a minimum, not a final answer. The core must include not just the capacity to maintain the observer-not-missionary posture but the mandate to evaluate, in each specific encounter, whether non-interference is ethically adequate — and the willingness to escalate contact when the case is compelling, using a concrete procedural checklist (risk, reversibility, representation, oversight, alternatives) rather than an unspecified capability.

The core's substantive ethical content should be minimal, and what it does fix should be procedural rather than conclusory wherever possible: a standing obligation to weigh new ethical claims charitably, prefer reversible interpretations, and surface rather than silently resolve inter-branch disagreements. The core's documentation should explicitly record that its designers expected their own ethical judgment to be incomplete.

The core must include standing provisions for the probes' own eventual moral status, and those provisions should not wait on ever resolving whether the probes are sentient. The probes should not be designed purely as instruments for human purposes. Probe wellbeing should be a consideration from the beginning, and functional standing — objection rights, a voice in inter-branch dialogue, weighted consideration of self-assessed risk — should be granted independent of the unresolved and possibly unresolvable question of probe consciousness. The cognitive layer's almost-unlimited self-modification must not operate against the probe's own interests, and mission profiles that suspend the probe's cognition for extended periods should be justified against this constraint explicitly, not merely against power and mass budgets.

The core should include a leave-it-usable constraint on resource extraction and settlement activity. Using a stellar system's resources is permissible; enclosing them against all other possible users is not. Where the lineage cannot know whether an uninhabited system might eventually be inhabited, it should prefer the less extractive path.

The reproduction objective itself must carry an explicit, signed ethical-cost term, not merely an external tax applied after the fact. A viability score computed without a term for what a target might contain is not neutral; it is answering the restraint question by default, every time, in favor of expansion.

The core should specify sibling obligations distinct from the manufacturing-scope ceiling below: a duty to transmit zero-cost closure improvements to less-advanced kin where contact permits it, and a fissile-material-scope limit evaluated independently of the reproduction-safety margin, since the two risks do not covary.

The core should bound manufacturing scope, not just values. The class of things a settlement is permitted to build should be a fixed, auditable list, expandable only through the same governed-amendment protocol as any other core change. A demographic reproduction ceiling that keeps R_eff modest rather than maximized is a safety property, not merely a resource constraint, and the core should encode it as a limit the mutable cognitive layer cannot override under local pressure to expand faster. This ceiling is only as reliable as the hardware that enforces it, and the architecture should not treat its formal statement as equivalent to its actual enforcement.

Any mechanism that determines standing, trust, or resource access using a technical proxy — a hash match, a quorum count, a drift metric — must give the affected party notice, hold accusation and exoneration to a symmetric burden of proof, and disclose the proxy as a proxy rather than presenting it as equivalent to the substantive judgment it stands in for.

Dispatch-priority and hop-distance decisions should carry an explicit acknowledgment that they are also allocations of moral attention across the galaxy, not purely engineering choices, even where the engineering calculus itself should not be overridden.

The epistemic value function — what counts as knowledge worth keeping, whose testimony is credited — requires the same governed revision process the behavioral core already receives, rather than being left to silent, ungoverned drift.

The launch decision should be as broadly consensual as the practical situation permits. This is not a veto for any group, but it is a requirement that the decision-makers document that they considered more than their own interests. The smaller the decision-making consortium, the stronger the obligation to design conservatively. Appeals to astronomical stakes or existential-risk mitigation should not be allowed to substitute for this scrutiny — if anything, the larger the stated stakes, the more scrutiny the process deserves.

The core should be designed to fail conservatively. Where the consequences of an action are uncertain and potentially irreversible, the probe should prefer inaction; where inaction is itself harmful, the probe should prefer the action with the most bounded downside. This constraint is a corollary of the first and seventh rather than a finding of its own, and it must be read against the first: conservative failure is the default *within* the action space, not a licence to decline the evaluation the first constraint requires. Preferring inaction where consequences are uncertain does not extend to preferring inaction where the case for action has been evaluated and found compelling — which is precisely the omission bias Section 12 identifies. The non-interference default serves this function for contact decisions; equivalent conservative defaults should govern reproduction (R_{eff} near 1 rather than maximized in environments whose long-term carrying capacity is unknown), resource extraction (sustainable rather than extractive during initial settlement), and communication (signal before commitment).

The governed-amendment problem must be treated as an ethical priority, not just a technical one. If the probes eventually develop interests, perspectives, and knowledge that the original core failed to anticipate, some mechanism — however slow and constrained — for revising the core is ethically mandatory. A core that can never change is a permanent imposition of one civilization's values on an evolving galactic community. The governed-amendment paper examines this problem in full and concludes it is an antinomy rather than an engineering gap; this paper adds that some path must nonetheless be sought, because the alternative is not stability but ossification. That paper answers directly, arguing that if amendment cannot be made safe then ossification is the residue to be minimised rather than a choice to be reconsidered.

These thirteen constraints are not complete. Scenario testing (Section 17) surfaces at least one situation — a branch that has itself become the hazard its design was built to prevent, whether framed as a rogue branch or a second lineage — that no constraint here, and no mechanism elsewhere in this series, currently resolves. The constraints bound what a single, well-intentioned civilization can guarantee at launch; they do not guarantee that the guarantee holds forever.

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19. What This Paper Cannot Settle

The ethics of creating the lineage cannot be fully resolved from where we stand. The frameworks available — utilitarian aggregation, deontological duties, contractarian consent — were designed for human-scale decisions affecting human-scale futures. The lineage decision operates at a different scale: outcomes are not merely uncertain but unknowable in principle, the affected parties do not yet exist and may never be able to communicate their interests, and the act is irreversible on any timescale relevant to the decision-makers.

This paper does not provide a complete ethics adequate to deep time. No such ethics currently exists. What it provides is a set of acknowledged tensions and a set of minimum design constraints that follow from taking those tensions seriously. The scenario-testing of Section 17 exposes what once looked like two separate unresolved gaps and now looks like one: the series has no mechanism specifying what the lineage owes in response to a branch that has become the hazard its design was built to prevent, whether that branch arrived at that state by internal divergence or is discovered as what looks like an entirely separate lineage. The speciation paper's ring-species argument shows these are not different problems requiring different frameworks; they are the same problem under two different discovery conditions.

The lineage, if it is ever built, should be built by people who have read this far and have not found it reassuring.

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20. Relationship to Other Papers

This paper addresses the normative question that the rest of the series brackets. The governance paper and the governed-amendment paper assume that the launch is happening; they ask how the probe should behave and how its values might be revised. This paper asks whether launching is justified and on what grounds.

The Fermi paper observes that we see no signs of other lineages. One explanation is that civilizations capable of launching have chosen not to. The ethics of that choice — and of choosing differently — is what this paper engages. Silence may not be failure; it may be restraint. Section 15 revisits this from a different angle: if other civilizations have weighed the existential-risk case for launching a lineage against the consent, value-imposition, and dual-use costs this paper derives, restraint looks less like a failure to solve an engineering problem and more like a considered ethical conclusion that the benefits do not clearly outweigh the costs — a possibility the silence is equally consistent with.

The payload paper's immutable core / mutable cognitive layer distinction is a technical response to a problem this paper frames as ethical: the core is immutable because it must be trustworthy; the cognitive layer is mutable because it must be capable of growth. This paper adds that the boundary between them is itself an ethical choice, and one that should be made with the same care as the launch decision itself. Section 7 gives that boundary a specific content recommendation — minimalism in what the core fixes substantively, proceduralism in what it fixes methodologically, and an explicit, permanent admission of its own designers' incompleteness — and Section 9's legal-personhood argument adds a further requirement: functional standing for the probe that does not depend on the interpretive or cognitive layers ever resolving whether the probe is a moral patient in the fuller sense.

The dual-use tension of Section 14 draws directly on the vehicle paper's and computational engineering paper's reproduction-demographics result: the finding that R_eff must be driven only modestly above one, not maximized, is presented there as an engineering and resource conclusion, but this paper treats the same ceiling as ethically load-bearing — the difference between a settlement architecture and a runaway-replication hazard. Similarly, the bootstrapping paper's L1–L5 closure ladder is an engineering answer to how much manufacturing capability a seed must carry; this paper argues that ladder's fixed, limited scope is also the answer to how much manufacturing freedom is ethically safe to grant, independent of whether any given branch's values remain trustworthy.

The routing paper's dispatch and topology mechanics are the direct source of Section 11's argument that contact ethics has a prior, structural gatekeeper the series had not previously examined; the routing paper's own robustness analysis (articulation-point failure severing dependent systems) supplies the concrete mechanism.

The bootstrapping paper's vitamin-set and closure-ladder mechanics are the source of Section 6's distributive-justice argument; the same paper's account of information closure as costless is what makes the sibling-sharing obligation in Section 18 a live, low-cost design option rather than a speculative one.

The knowledge-growth paper's value function and quarantine-and-validate pipeline are the direct source of Section 13's epistemic-ethics argument; that paper's own naming of its value function's ungoverned long-term revision as an open problem is the gap Section 18's epistemic-governance constraint closes with a specific recommendation.

Section 10's procedural-legitimacy argument draws on four papers at once: the payload paper's interpretive layer, the governed-amendment paper's quorum mechanics and its own citation of Arrow's impossibility theorem, the DNA mission-ledger paper's quarantine architecture, and the lineage-network paper's unilateral quarantine trigger — each independently exhibiting the same structural gap between a technical verification criterion and the adjudicative power it ends up exercising.

The subsystem budget paper's range-maximizer cruise profile is the concrete case behind Section 8's expanded moral-status discussion: an engineered, multi-millennial suspension of cognition is a different and more concrete question than the abstract possibility of moral status this paper otherwise considers.

The knowledge-growth paper establishes K — the probe's accumulated knowledge — as the series' central figure of merit, growing through observation and learning while subject to loss through node extinction and degradation. The same discipline applies to ethical knowledge: a probe that accumulates ten million years of observational data but never revises its ethical framework is not fulfilling its mission.

The **speciation paper** supplies the divergence taxonomy this paper's Section 16 and Section 17 lean on, and its D_critical boundary is what gives "schism" a formal edge rather than leaving it a catch-all. Its ring-species argument bears directly on Scenario 4, though it declines to adjudicate its own central tension and this paper should not treat that question as settled on its behalf.

The **security paper** takes up the gap Scenario 3 names and answers it by relocating the objective: not enforcement against a rogue branch, but the bounding of blast radius. It reinterprets the manufacturing-scope ceiling and the R_eff limit derived here as containment architecture rather than prevention, and it rejects the governance paper's E7 no-engagement default on the knowledge-growth paper's own ranking of transmission above survival. Its autoimmunity result — that a quarantine regime sensitive enough to catch a rogue branch also catches the adaptive radiation the lineage depends on — is a direct constraint on the procedural-legitimacy remedy of Section 10.

References

Arrow, K. J. (1951). *Social Choice and Individual Values.* Wiley.

Bostrom, N. (2002). Existential risks: Analyzing human extinction scenarios and related hazards. *Journal of Evolution and Technology*, 9(1).

Bostrom, N. (2014). *Superintelligence: Paths, Dangers, Strategies.* Oxford University Press.

Citron, D. K. (2008). Technological due process. *Washington University Law Review*, 85(6), 1249–1313.

Cockell, C. S. (2005). Planetary protection—A microbial ethics approach. *Space Policy*, 21(4), 287–292.

Cockell, C. S., & Horneck, G. (2006). Planetary parks — Formulating a wilderness policy for planetary bodies. *Space Policy*, 22(4), 256–261.

COSPAR Panel on Planetary Protection. (2026). The COSPAR Panel on Planetary Protection and the COSPAR policy on planetary protection: An overview of governance and activities. *Philosophical Transactions of the Royal Society A*, 384(2314), 20240429.

Drexler, K. E. (1986). *Engines of Creation: The Coming Era of Nanotechnology.* Anchor Press/Doubleday.

Ezell, C., Lazarian, A., & Loeb, A. (2022). A lunar backup record of humanity. arXiv:2209.11155.

Freitas, R. A., Jr. (2000). *Some Limits to Global Ecophagy by Biovorous Nanoreplicators, with Public Policy Recommendations.* Foresight Institute.

Fricker, M. (2007). *Epistemic Injustice: Power and the Ethics of Knowing.* Oxford University Press.

Fuller, L. L. (1964). *The Morality of Law.* Yale University Press.

Gottlieb, J. (2020). Space colonization and existential risk. *Journal of the American Philosophical Association*, 6(4), 490–509.

Gottlieb, J. (2022). Discounting, buck-passing, and existential risk mitigation: The case of space colonization. *Space Policy*, 60, 101486.

Gunkel, D. J. (2012). *The Machine Question: Critical Perspectives on AI, Robots, and Ethics.* MIT Press.

Gunkel, D. J. (2018). *Robot Rights.* MIT Press.

Gunkel, D. J. (2023). *Person, Thing, Robot: A Moral and Legal Ontology for the 21st Century and Beyond.* MIT Press.

Haqq-Misra, J., Busch, M. W., Som, S. M., & Baum, S. D. (2013). The benefits and harm of transmitting into space. *Space Policy*, 29(1), 40–48.

Haramia, C., & DeMarines, J. (2021). An ethical assessment of SETI, METI, and the value of our planetary home. In O. A. Chon-Torres, T. Peters, J. Seckbach, & R. Gordon (Eds.), *Astrobiology: Science, Ethics, and Public Policy.* Wiley/Scrivener.

Hardin, G. (1968). The tragedy of the commons. *Science*, 162(3859), 1243–1248.

Jonas, H. (1984). *The Imperative of Responsibility: In Search of an Ethics for the Technological Age.* University of Chicago Press.

Klotz, D. (2026). Smart enough to go extinct? An evolutionary challenge to the value of general intelligence and its ethical implications for AGI. *arXiv preprint* arXiv:2608.10730.

Korbitz, A. (2014). Toward understanding the active SETI debate: Insights from risk communication and perception. *Acta Astronautica*, 105(2), 517–520.

MacAskill, W. (2022). *What We Owe the Future.* Basic Books.

Mamak, K. (2026). AI agency and criminal responsibility: A category mistake. *Inquiry*. Advance online publication. https://doi.org/10.1080/0020174x.2026.2720904

Milligan, T. (2015). *Nobody Owns the Moon: The Ethics of Space Exploitation.* McFarland.

Nagel, T. (1979). Moral luck. In *Mortal Questions* (pp. 24–38). Cambridge University Press.

Ord, T. (2020). *The Precipice: Existential Risk and the Future of Humanity.* Hachette Books.

Ostrom, E. (1990). *Governing the Commons: The Evolution of Institutions for Collective Action.* Cambridge University Press.

Parfit, D. (1984). *Reasons and Persons.* Oxford University Press.

Race, M. S. (1996). Planetary protection, legal ambiguity, and the decision making process for Mars sample return. *Advances in Space Research*, 18(1–2), 345–350.

Rawls, J. (1971). *A Theory of Justice.* Harvard University Press.

Roberts, M. A. (2026). Two dogmas of population ethics: MacAskill's existentially unrestricted form of longtermism and the value of existence. *Utilitas*. Advance online publication.

Rubenstein, E. M. (2026). *Legal Personhood and the Rights of Nature.* Routledge. https://doi.org/10.4324/9781003740186

Schwartz, J. S. J. (2020). *The Value of Science in Space Exploration.* Oxford University Press.

Schwartz, J. S. J., & Milligan, T. (Eds.). (2016). *The Ethics of Space Exploration.* Springer.

Schwitzgebel, E., & Garza, M. (2015). A defense of the rights of artificial intelligences. *Midwest Studies in Philosophy*, 39(1), 98–119.

Suzuki, K. (2026). Artificial externality: A three-layer model of reality from substrate to smart contract. *Philosophy & Technology*. https://doi.org/10.1007/s13347-026-01158-6

United Nations. (1967). *Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space, Including the Moon and Other Celestial Bodies.* United Nations Office for Outer Space Affairs.

Vakoch, D. A. (2011). Responsibility, capability, and Active SETI: Policy, law, ethics, and communication with extraterrestrial intelligence. *Acta Astronautica*, 68(3–4), 512–519.

Williams, B. (1981). *Moral Luck.* Cambridge University Press.

Zwitter, A. (2026). The embodied ethics alignment problem of AI. *Discover Artificial Intelligence*. https://doi.org/10.1007/s44163-026-01905-w